	Scotch Student ID #			
Circle the relevant digits	0	0	0	0
	1	1	1	1
	2	2	2	2
	3	3	3	3
	4	4	4	4
	5	5	5	5
	6	6	6	6
	7	7	7	7
	8	8	8	8
	9	9	9	9

Teacher's Name

Scotch College

CHEMISTRY

SAC 2 – RATES, EQUILIBRIUM & REDOX

Date: Wednesday, May 26th

Reading Time	0 minutes
Writing Time	45 minutes

Task Sections	Marks	Your Marks
Total Marks	41	

General Instructions

- Write your student number in the space provided above on this page.
- Please complete all questions in the spaces provided.

Allowed Materials

- A VCAA Chemistry Data Book has been provided.
- Students are permitted to bring into the SAC: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator.

At the end of the task…

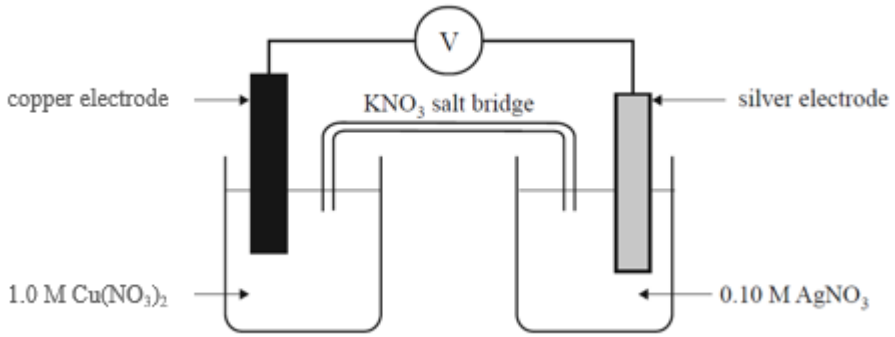
- Please ensure your student number has been filled in at the top of the page and that you have written your teacher's name.

Electronic Devices

Students are **not** allowed to have a mobile phone, smart watch and/or any other unauthorised electronic device in the SAC, unless it is TURNED OFF and is placed on the front teacher desk.

Q1. *(1 mark)*

Consider the following electrochemical cell.



Which one of the following is correct?

As the cell discharges:

	electrons would flow from the	**in the salt bridge**
A.	copper electrode to the silver electrode.	anions migrate to the Cu^{2+}/Cu half-cell.
B.	silver electrode to the copper electrode.	cations migrate to the Cu^{2+}/Cu half-cell.
C.	silver electrode to the copper electrode.	cations migrate to the Ag^{+}/Ag half-cell.
D.	copper electrode to the silver electrode.	anions migrate to the Ag^{+}/Ag half-cell.

Q2. *(1 mark)*

Which factors affect the amount of product formed at the cathode during electrolysis of molten salts?

I. current II. pH of the cell III. charge on the cation

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

Q3. *(1 mark)*

For $2N_2O\ (g) \rightleftharpoons 2N_2\ (g) + O_2\ (g)$ Kc = 7.3×10^{34} M.

What is Kc for the following reaction, at the same temperature?

$N_2\ (g) + \frac{1}{2}O_2\ (g) \rightleftharpoons N_2O\ (g)$

A. 7.3×10^{34}

B. $\frac{1}{2\times7.3\times10^{34}}$

C. $\frac{2}{7.3\times10^{34}}$

D. $\frac{1}{\sqrt{7.3\times10^{34}}}$

Q4. *(9 marks)*

A mixture of 1.20 mol sulfur dioxide, SO_2 (g), 2.40 mol oxygen gas, O_2 (g) and 1.20 mol sulfur trioxide, SO_3 (g) is placed in a 1.00 L container and allowed to reach equilibrium at temperature T.

$$2SO_2\,(g) + O_2\,(g) \rightleftharpoons 2SO_3\,(g)\ \Delta H = -296.8\ \mathrm{kJ/mol}$$

a. Distinguish between the terms reaction quotient (concentration fraction), *Q,* and equilibrium constant, Kc. (2 marks)

b. The equilibrium constant, Kc, is 0.382 M^{-1} at temperature T. Deduce, showing your work, the direction (back or forward) of the initial reaction. (2 marks)

c. Another equilibrium mixture was found to contain concentrations of 0.140 M SO_2 and 0.180 M oxygen gas, O_2 when the Kc is 1.75 M^{-1}. Calculate the equilibrium concentration of SO_3. (2 marks)

$$2SO_2\,(g) + O_2\,(g) \rightleftharpoons 2SO_3\,(g)\ \Delta H = -296.8\ \mathrm{kJ/mol}$$

d. Predict how each of the following changes affects the position of equilibrium and the value of Kc. You may use arrows, → ← ↑ ↓ (or 'no effect') to show your answer. (3 marks)

	Position of equilibrium (right, left or no effect)	**Value of Kc (up, down or no effect)**
Addition of a catalyst		
Increase in pressure by decreasing volume		
Temperature is decreased		

Q5. *(3 marks)*

Alcohols can be oxidized with acidified potassium dichromate solution to form a variety of compounds including carboxylic acids. The balanced overall equation for one of these reactions is:

$$3CH_3OH\ (l) + 2Cr_2O_7^{2-}\ (aq) + 16H^+\ (aq) \rightleftharpoons 3CHOOH\ (aq) + 4Cr^{3+}\ (aq) + 11H_2O\ (l)$$

a. Identify the oxidising agent. Give your reasoning using oxidation numbers. (2 marks)

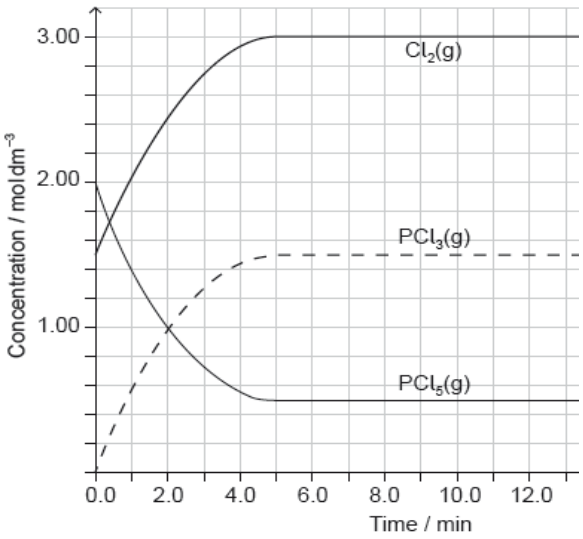
b. Deduce the reduction half-equation for this reaction. States are not required. (1 mark)

Q6. (10 marks)

PCl_5 (g) decomposes according to the following equation:

$$PCl_5\ (g) \rightleftharpoons PCl_3\ (g) + Cl_2\ (g) \quad \Delta H = +87.8\ \mathrm{kJ/mol}$$

PCl_5 (g) and Cl_2 (g) were placed in a sealed flask and allowed to reach equilibrium at 200 °C.



[Source: http://education.alberta.ca/media]

a. Calculate the value of the equilibrium constant, Kc. (2 marks)

b. For the following **two** characteristics of a dynamic equilibrium, fill in the blank left after each statement to make them correct. (2 marks)

When in a state of dynamic equilibrium:

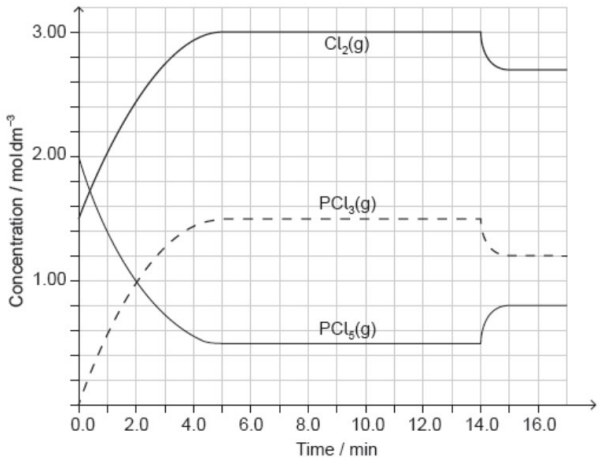
i. the rates of the forward and back reactions will

_____.

ii. the concentrations of products and reactants will

_____.

$$PCl_5\,(g) \rightleftharpoons PCl_3\,(g) + Cl_2\,(g) \quad \Delta H = +87.8\ \mathrm{kJ/mol}$$



[Source: http://education.alberta.ca/media]

c. Identify the **specific** change responsible for the equilibrium shift observed after 14 minutes. Explain your answer. (2 mark)

_____

_____

_____

d. How does each of the following changes affect the amount of PCl_3 in the equilibrium mixture? In your answers, refer to Le Chatelier's principle. (4 marks)

i. The volume of the container is increased.

_____

_____

_____

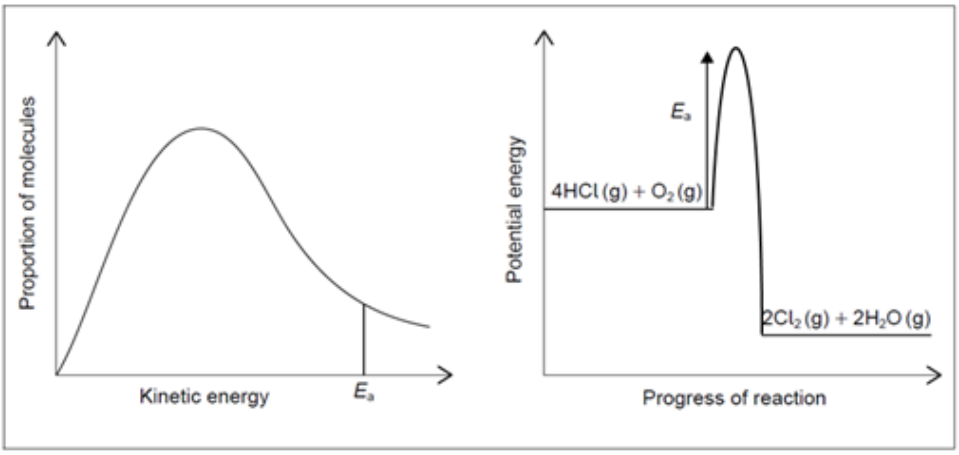
ii. PCl_5 is added to the equilibrium mixture.

Q7. (12 marks)

The diagram shows the Maxwell–Boltzmann distribution and potential energy profile for the following reaction without a catalyst.

$$4HCl\ (g) + O_2\ (g) \rightleftharpoons 2Cl_2\ (g) + 2H_2O\ (g)$$

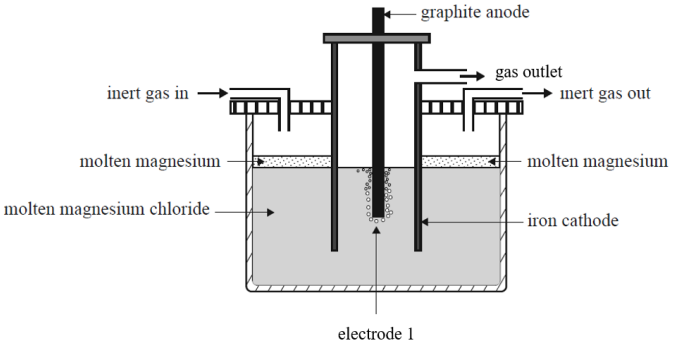
a. Annotate **both** charts to show the activation energy for the catalysed reaction, using the label $\boldsymbol{E_{a\ (cat)}}$.



(2 marks)

b. Explain in general terms how the catalyst increases the rate of the reaction. (1 mark)

An electrolysis cell was assembled using a graphite electrode and an iron electrode and connected as shown. Inert gas flows through the cell to keep the pressure constant within the cell.



c. Write the half-equation for the formation of gas bubbles at electrode 1. (2 marks)

d. Explain why the graphite electrode cannot be replaced with another iron electrode. (2 marks)

e. Using the electrochemical series, determine the applied voltage required for the electrolysis cell to operate. (1 mark)

f. If the cell ran for 2.00 minutes with a current of 1.20 A, calculate the volume of gas that would form at SLC. (4 marks)

Q8. (4 marks)

A reactivity series can be experimentally determined by adding the metals W, X, Y and Z to solutions of these metal ions. The following spontaneous reactions were observed:

$$Y^{2+}(aq) + Z(s) \rightarrow Y(s) + Z^{2+}(aq)$$

$$X(s) + Y^{2+}(aq) \rightarrow X^{2+}(aq) + Y(s)$$

$$W^{2+}(aq) + Y(s) \rightarrow W(s) + Y^{2+}(aq)$$

$$X(s) + Z^{2+}(aq) \rightarrow X^{2+}(aq) + Z(s)$$

a. Deduce the order of reactivity of these four metals, from the weakest reductant to the strongest reductant. (1 mark)

b. A secondary cell is made by connecting a half-cell of W in W^{2+} (aq) to a half-cell of Z in Z^{2+} (aq). Deduce the overall equation for the reaction taking place when the cell is **discharging**. (1 mark)

c. One of these cells was constructed and kept next to a heat source. Describe how this might affect the performance of this secondary cell over its lifetime. (2 marks)

End of SAC